

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the **A Comprehensive Measure of Well-Being (HDI Predictor)** project. Each feature contributes to building an intelligent Human Development Index (HDI) prediction system using Machine Learning and a Flask web application.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collects Human Development Index dataset containing health, education, and income indicators.	Dataset Module	Done	4
2	Data Preprocessing	Cleans dataset, handles missing values, feature selection, and data scaling.	Data Processing	Done	6
3	Data Visualization	Generates charts and visual analysis for HDI indicators and relationships.	Visualization Module	Done	5
4	Machine Learning Model	Trains and evaluates the HDI prediction model using Machine Learning algorithms.	ML Module	Done	12

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
5	HDI Prediction	Predicts the Human Development Index based on user input values.	Prediction Engine	Done	8
6	Flask Web Application	Provides a responsive web interface for entering data and viewing predictions.	Flask Application	Done	7
7	Prediction Result Display	Displays predicted HDI value along with interpretation of development level.	Result Module	Done	4
8	Model Deployment	Project prepared for local deployment and future cloud deployment.	Deployment Module	Pending	4

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Prediction Form & Result Page
2	Backend / Logic	3	Data Preprocessing, ML Prediction, Flask Routing
3	Database / Storage	1	HDI Dataset (CSV) and Trained Model (.pkl)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	0	Not Implemented in Current Version